Question # 1

1

a) $x_1 + 2x_2 - 3x_3 = 1$

$2x_1 + x_2 - 3x_3 = 0$

$-x_1 + x_2 = -1$

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 1 \\ 2 & 1 & -3 & 0 \\ -1 & 1 & 0 & -1 \end{array}\right] \overset{R}{\sim} \left[\begin{array}{ccc|c} 1 & 2 & -3 & 1 \\ 0 & -3 & 3 & -2 \\ 0 & 3 & -3 & 0 \end{array}\right] \begin{array}{l} \\ R_2 - 2R_1 \\ R_3 + R_1 \end{array}$$

$$\overset{R}{\sim} \left[\begin{array}{ccc|c} 1 & 2 & -3 & 1 \\ 0 & -3 & 3 & -2 \\ 0 & 0 & 0 & -2 \end{array}\right] \begin{array}{l} \\ \\ R_3 + R_2 \end{array}$$

Since $0 \neq -2$

the system is inconsistent.

We have no solution

b)
$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 2 & 1 & -3 & 0 \\ -1 & 1 & 0 & 0 \end{array}\right]$$

$$\overset{R}{\sim} \left[\begin{array}{ccc|c} 1 & 2 & -3 & 0 \\ 0 & -3 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

x_3 is free varieble.

Suppose $x_3 = s$

$x_1 + 2x_2 - 3x_3 = 0$ — (1)

$-3x_2 + 3x_3 = 0$ — (2)

From eq (2)

$3x_2 = 3x_3$

$x_2 = x_3 = s$.

From eq (1)

$x_1 = -2x_2 + 3x_3$

$= -2s + 3s$

$x_1 = s$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} s$$

c) The solution of $A(x) = b$ is generally a transformat-ion (translation) of $A(x) = 0$

In this case since the solution of $A(x) = b$ does not exist, there is no relation.

(٢)

Question 2

a)

→ If there are less then 3 vectors, span of these vectors will not be R^3 rather a subspace of R^3. So the condition that basis should span the whole space R^3 is not satisfied.

→ If there are more then 3 vectors they will not be linearly independent hence the condition that basis vectors must be linearly independent is not satisfied.

b) There are four pivot entries so Rank = 4

By Rank Theorem : Rank + Nullity = Dimension (# of columns)

$$4 + \text{Nullity} = 6$$

$$\text{Nullity} = 6 - 4 = 2$$

c) $\begin{bmatrix} 1 & 4 & 8 & -3 & -7 \\ -1 & 2 & 7 & 3 & 4 \\ -2 & 2 & 9 & 5 & 5 \\ 3 & 6 & 9 & -5 & -2 \end{bmatrix}$

$$\overset{R}{\sim}\begin{bmatrix} 1 & 9 & 8 & -3 & -7 \\ 0 & 6 & 15 & 0 & -3 \\ 0 & 10 & 25 & -1 & -9 \\ 0 & -6 & -15 & 4 & 19 \end{bmatrix} \begin{matrix} \\ R_2+R_1 \\ R_3+2R_1 \\ R_4-3R_1 \end{matrix}$$

$$\overset{R}{\sim}\begin{bmatrix} 1 & 9 & 8 & -3 & -7 \\ 0 & 1 & 15/6 & 0 & -1/2 \\ 0 & 10 & 25 & -1 & -9 \\ 0 & -6 & -15 & 4 & 19 \end{bmatrix} \begin{matrix} \\ R_2/6 \\ \\ \\ \end{matrix}$$

$$\overset{R}{\sim}\begin{bmatrix} 1 & 9 & 8 & -3 & -7 \\ 0 & 1 & 15/6 & 0 & -1/2 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 4 & 16 \end{bmatrix} \begin{matrix} \\ \\ R_3-10R_2 \\ R_4+6R_2 \end{matrix}$$

$$\overset{R}{\sim}\begin{bmatrix} \boxed{1} & 9 & 8 & -3 & -7 \\ 0 & \boxed{1} & 5/2 & 0 & -1/2 \\ 0 & 0 & 0 & \boxed{1} & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ \\ \\ R_4-4R_3 \end{matrix}$$

Pivot Columns are v_1 v_2 and v_2

$$\text{Column space} = \left\{ \begin{bmatrix} 1 \\ -1 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 2 \\ 2 \\ 6 \end{bmatrix}, \begin{bmatrix} -3 \\ 3 \\ 5 \\ -5 \end{bmatrix} \right\}$$

<u>Question 3</u> A transpormed $T: R^a \to R^b$ is defined from

$R^{\text{columns}} \longrightarrow R^{\text{row}}$

$a = 3$ $b = 2$

$R^3 \longrightarrow R^2$

(b) $T: \mathbb{R}^2 \to \mathbb{R}^2$. (4)

T_1 = Horizontal shear = $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$.

T_2 = Reflection about $y=-x \Rightarrow T_2 = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$.

T_3 = Rotation by $\pi/2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

Then $T\begin{pmatrix} x \\ y \end{pmatrix} = T_3 \circ T_2 \circ T_1 \begin{pmatrix} x \\ y \end{pmatrix}$.

$$= \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}.$$

(c) Given $T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 & 7 \\ 2 & 8 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$

$$\left(\begin{array}{cc|c} 1 & 7 & 0 \\ 2 & 8 & 0 \\ 3 & 4 & 0 \end{array}\right) \underset{R_3-3R_1}{\overset{R_2-2R_1}{\sim}} \left(\begin{array}{cc|c} 1 & 7 & 0 \\ 0 & -6 & 0 \\ 0 & -17 & 0 \end{array}\right) \overset{-1/6 R_2}{\sim} \left(\begin{array}{cc|c} 1 & 7 & 0 \\ 0 & 1 & 0 \\ 0 & -17 & 0 \end{array}\right)$$

$$\overset{R_3+17R_2}{\sim} \left(\begin{array}{cc|c} 1 & 7 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right)$$

Now, since we have two pivots

$\Rightarrow$ Rank $= 2 \Rightarrow$ Range = Span $\left\{\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 7 \\ 8 \\ 4 \end{pmatrix}\right\}$

By Fundamental Theorem of Linear algebra

Rank + Nullity $= 2 \Rightarrow 2 +$ Nullity $= 2$

$\Rightarrow$ Nullity $= 0$

$\Rightarrow$ Null$(T) = \{0\}$.

(d) $T\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -5x_1 + 9x_2 \\ 4x_1 - 7x_2 \end{pmatrix}$.

$$T\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -5 & 9 \\ 4 & -7 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Thus, $T^{-1}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -5 & 9 \\ 4 & -7 \end{pmatrix}^{-1}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

Now, $\begin{pmatrix} -5 & 9 \\ 4 & -7 \end{pmatrix}^{-1} = \frac{1}{35-36}\begin{pmatrix} -7 & -9 \\ -4 & -5 \end{pmatrix} = \begin{pmatrix} 7 & 9 \\ 4 & 5 \end{pmatrix}$.

$\Rightarrow T^{-1}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 7 & 9 \\ 4 & 5 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 7x_1 + 9x_2 \\ 4x_1 + 5x_2 \end{pmatrix}$.

Question 4: $T : \mathbb{R}^2 \longrightarrow \mathbb{R}^3$, defined as · (6)

$$T\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -5x_1 + 9x_2 \\ 4x_1 - 7x_2 \\ x_1 + 3x_2 \end{pmatrix}.$$

(a) Matrix of $T = [T] = \begin{pmatrix} -5 & 9 \\ 4 & -7 \\ 1 & 3 \end{pmatrix}$.

(c) To prove T is one-one.

Consider $\left(\begin{array}{cc|c} -5 & 9 & 0 \\ 4 & -7 & 0 \\ 1 & 3 & 0 \end{array}\right) \overset{R_1 \leftrightarrow R_3}{\sim} \left(\begin{array}{cc|c} 1 & 3 & 0 \\ 4 & -7 & 0 \\ -5 & 9 & 0 \end{array}\right)$

$\underset{R_3+5R_1}{\overset{R_2-4R_1}{\sim}} \left(\begin{array}{cc|c} 1 & 3 & 0 \\ 0 & -19 & 0 \\ 0 & 24 & 0 \end{array}\right) \underset{1/24 R_3}{\overset{-1/19 R_2}{\sim}} \left(\begin{array}{cc|c} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{array}\right) \overset{R_3-R_2}{\sim} \left(\begin{array}{cc|c} 1 & 3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right).$

Since we have two pivots $\Rightarrow$ Rank $[T] = 2$.

$$\text{Rank}[T] + \text{Nullity}[T] = 2.$$

$$2 + \text{Nullity}[T] = 2$$

$$\text{Nullity}[T] = 0$$

$\Rightarrow$ Null $([T]) = 0 \Rightarrow \ker T = 0$ & so, T is one-one.

(b) From (c) we observe that, $\text{Rang}\, T = \text{span}\left\{\begin{pmatrix} -5 \\ 4 \\ 1 \end{pmatrix}, \begin{pmatrix} 9 \\ -7 \\ 3 \end{pmatrix}\right\}$ which is two dimensional

& $\dim \mathbb{R}^3 = 3$

$\Rightarrow \dim \text{Range}[T] \neq \dim \mathbb{R}^3$

& so, T cannot be onto.